

Experiment No. (3):

- a) Write a Java program to list all the files in a directory including the files present in all its subdirectories.

```
package recordprogs;

import java.util.*;
import java.io.*;

class FileDisplay
{
    public static void main(String args[])
    {
        String dirname;

        Scanner in = new Scanner(System.in);

        System.out.println("Enter directory name:");
        dirname = in.nextLine();

        listf(dirname);

        in.close();
    }

    public static List<File> listf(String directoryName)
    {
        File directory = new File(directoryName);
        List<File> resultList = new ArrayList<File>();

        // Check whether the directory exists
        if (!directory.exists())
        {
            System.out.println("Directory does not exist.");
            return resultList;
        }

        // Check whether it is a directory
        if (!directory.isDirectory())
        {
            System.out.println("The given path is not a directory.");
            return resultList;
        }

        // Get all files and directories
```

```

File[] fList = directory.listFiles();

if (fList == null)
{
    System.out.println("Unable to access the directory.");
    return resultList;
}

for (File file : fList)
{
    if (file.isFile())
    {
        System.out.println(file.getAbsolutePath());
        resultList.add(file);
    }
    else if (file.isDirectory())
    {
        resultList.addAll(listf(file.getAbsolutePath()));
    }
}

return resultList;
}
}

```

Output:

Enter directory name:

```

C:\Users\Administrator\Anvesh-workspace\II CSE G\src
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\animals\ProtectedAnimal.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\ChatBotServer.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\college\Student.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\FinalMethod.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\inheritance\Finalkeyword.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\inheritance\FinalMethod.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\inheritance\FinalVariable.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\inheritance\PolyMethodoverloading.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\inheritance\TestInheritance.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\inheritance\TestInheritance2.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\inheritance\TestInheritance3.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\interfaces\InterfaceDemo.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\interfaces\InterfaceImplements.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\interfaces\Interfaceinheritance.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\interfaces\MultipleInheritance.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\miniprojects\ACELibraryInventory.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\miniprojects\SmartGridAutomation.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Abstractclass.java

```

C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\AbstractMethod.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Addition.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Arithmetic.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\AssignmentOperator.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\BitwiseOperatorsDemo.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\BreakExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\BreakLoop.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Calculatorreturn.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\CallByReferenceExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\CallByValueExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\CarRacingGame.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\ChatBotServer.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\ConditionalOperator.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\ConstructorOverloadingDemo.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\ConstructorOverloadingDemo2.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\ContinueExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\DefaultConstructorDemo.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\DefaultModifierStudent.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\DefaultModifierTest.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\demo.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Demomain.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Demomain1.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\DoWhileExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\enemy.jpg
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\EvenOdd.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Factorial.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Fibonacci.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\ForEach.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\ForEachExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\ForExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\GamePanel.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\GetClassMethod.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\GunFireGame.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\HashCodeExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\HelloWorld.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\IfCondition.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\IfElseExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\IfElseIfExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Increment.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\InfixExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\LeapYear.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\LocalInneclass.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\LogicalOp.java

C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Main.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\MethodInitialization.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\MethodInstance.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\MethodOverLoading.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\MethodOverriding.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\MethodStatic.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\MethodStaticBlock.java
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G\src\oppsJavaProg\Methodstaticmethod.java
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G\src\oppsJavaProg\Methodstudentstatic.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\MethodUserdefined.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\MethodVoid.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\MethodWithoutParameters.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\MethodWithParameters.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\MethodWithReturnValue.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\NarrowingTypeCastingExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Nested.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\ObjecReference2.java
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C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\ObjectMethod.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\ObjectMethodExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Onedarray.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Operators.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\player.jpg
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\PolyMethodoverloading.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\PolyOverriding.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Prime.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\PrivateModifier.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\ReadValues.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\RecursionMethod.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\RelationalOperator.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\ReverseNumber.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Screenshot.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\SnakeGame.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Staticnestedclass.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Student6.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Superconstructor.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Supermethod.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Supervariable.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Swap.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\SwitchDemo.java
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G\src\oppsJavaProg\TestAnonymousInner.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\TestMemberOuter.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\TestStudent4.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\ThisConstructorChain.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\ThisCurrentClassMethod.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\ThisCurrentobject.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\ThisInstancevariables.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\ThisReturningobject.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\ThreeDArrayExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\Twodarray.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\TypePromote.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\oppsJavaProg\WhileExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE
G\src\oppsJavaProg\WideningTypeCastingExample.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\pack1\PubliceModifierStudent.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\pack2\PublicModifierTest.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\pets\ProctetedModifier.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\pets\ProtectedPets.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\recordprogs\AreaRTC.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\recordprogs\FileDisplay.java
C:\Users\Administrator\Anvesh-workspace\II CSE G\src\test\TestStudent.java

Viva Questions:

1. What is Java List Interface?

List Interface is the subinterface of Collection. It contains methods to insert and delete elements in index basis. It is a factory of ListIterator interface.

List Interface declaration

```
public interface List<E> extends Collection<E>
```

Methods of Java List Interface

Method	Description
void add(int index, Object element)	It is used to insert element into the invoking list at the index passed in the index.
boolean addAll(int index, Collection c)	It is used to insert all elements of c into the invoking list at the index passed in the index.
Object get(int index)	It is used to return the object stored at the specified index within the invoking collection.
Object set(int index, Object element)	It is used to assign element to the location specified by index within the invoking list.
Object remove(int index)	It is used to remove the element at position index from the invoking list and return the deleted element.
ListIterator listIterator()	It is used to return an iterator to the start of the invoking list.
ListIterator listIterator(int index)	It is used to return an iterator to the invoking list that begins at the specified index.

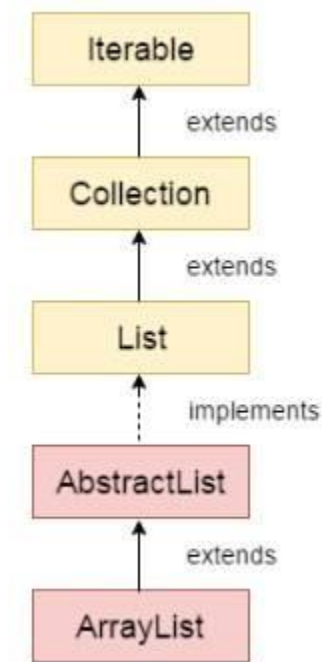
2. What is ArrayList class?

Java ArrayList class uses a dynamic array for storing the elements. It inherits AbstractList class and implements List interface.

The important points about Java ArrayList class are:

- Java ArrayList class can contain duplicate elements.
- Java ArrayList class maintains insertion order.
- Java ArrayList class is non synchronized.
- Java ArrayList allows random access because array works at the index basis.
- In Java ArrayList class, manipulation is slow because a lot of shifting needs to be occurred if any element is removed from the array list.

Hierarchy of ArrayList class



As shown in above diagram, Java ArrayList class extends AbstractList class which implements List interface. The List interface extends Collection and Iterable interfaces in hierarchical order.

ArrayList class declaration

Let's see the declaration for java.util.ArrayList class.

```
public class ArrayList<E> extends AbstractList<E> implements List<E>, RandomAccess, Cloneable, Serializable
```

Constructors of Java ArrayList

Constructor	Description
ArrayList()	It is used to build an empty array list.
ArrayList(Collection c)	It is used to build an array list that is initialized with the elements of the collection c.
ArrayList(int capacity)	It is used to build an array list that has the specified initial capacity.

Methods of Java ArrayList

Method	Description
void add(int index, Objectelement)	It is used to insert the specified element at the specified position index in a list.
boolean addAll(Collection c)	It is used to append all of the elements in the specified collection to the end of this list, in the order that they are returned by the specified collection's iterator.
void clear()	It is used to remove all of the elements from this list.
int lastIndexOf(Object o)	It is used to return the index in this list of the last occurrence of the specified element, or -1 if the list does not contain this element.
Object[] toArray()	It is used to return an array containing all of the elements in this list in the correct order.
boolean add(Object o)	It is used to append the specified element to the end of a list.
boolean addAll(int index, Collection c)	It is used to insert all of the elements in the specified collection into this list, starting at the specified position.
Object clone()	It is used to return a shallow copy of an ArrayList.
int indexOf(Object o)	It is used to return the index in this list of the first occurrence of the specified element, or -1 if the List does not contain this element.
void trimToSize()	It is used to trim the capacity of this ArrayList instance to be the list's current size.

b) Write a Java program on Random Access File class to perform different read and write operations

```
package recordprogs;
import java.io.RandomAccessFile;
import java.io.IOException;

public class RandomAccessFileExample {
    public static void main(String[] args) {
        try {
            // Open a random access file in read/write mode
            RandomAccessFile file = new RandomAccessFile("example.txt",
"rw");

            // Write some data to the file
            file.writeUTF("Hello, world!");
            file.writeInt(42);
            file.writeBoolean(true);

            // Move the file pointer to the beginning of the file
            file.seek(0);

            // Read the data back from the file
            String str = file.readUTF();
            int num = file.readInt();
            boolean bool = file.readBoolean();

            // Print out the data that was read from the file
            System.out.println(str);
            System.out.println(num);
            System.out.println(bool);

            // Move the file pointer to a specific location and overwrite some data
            file.seek(12);
            file.writeUTF("Java");
```

```

// Move the file pointer to the end of the file and append some data
file.seek(file.length());
file.writeInt(99);

// Close the file
file.close();
} catch (IOException e) {
    e.printStackTrace();
}
}
}

```

OUTPUT:-

example.txt is created in **C:\Users\Administrator\Anvesh-workspace\II CSE G**

